THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS CHAPTER TEST SET B

Chapter 7: **The Mathematics of Maybe: Introduction to Probability** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A fair die is rolled once. The probability of getting a number greater than 4 is: 1

(a) $\frac{1}{2}$ **(b)** $\frac{1}{3}$ **(c)** $\frac{1}{4}$ **(d)** $\frac{2}{3}$

2. A fair die is rolled once. The probability of getting a number other than 6 is: 1

(a) $\frac{5}{7}$ **(b)** $\frac{5}{6}$ **(c)** $\frac{2}{3}$ **(d)** $\frac{1}{6}$

3. A bag contains 4 red, 6 blue and 2 green balls. One ball is drawn at random. The probability that it is red is: 1

(a) $\frac{2}{3}$ **(b)** $\frac{1}{4}$ **(c)** $\frac{1}{3}$ **(d)** $\frac{1}{2}$

4. A bag contains 2 red, 8 blue and 5 green balls. One ball is drawn at random. The probability that it is green is: 1

(a) $\frac{1}{3}$ **(b)** $\frac{2}{3}$ **(c)** $\frac{1}{4}$ **(d)** $\frac{1}{2}$

5. A letter is picked at random from the word ‘PROBABILITY’. The probability of picking a vowel is: 1

(a) $\frac{7}{11}$ **(b)** $\frac{4}{11}$ **(c)** $\frac{5}{11}$ **(d)** $\frac{1}{3}$

6. An event occurred 8 times in 50 trials. Its experimental probability (relative frequency) is: 1

(a) 6.25 **(b)** 0.08 **(c)** 0.26 **(d)** 0.16

7. An event occurred 28 times in 80 trials. Its experimental probability (relative frequency) is: 1

(a) 0.35 **(b)** 2.86 **(c)** 0.45 **(d)** 0.175

8. A sample shows a probability of $\frac{2}{5}$ that a student likes cricket. In a school of 1500 students the expected number who like cricket is: 1

(a) 750 **(b)** 700 **(c)** 600 **(d)** 300

9. The probability of an impossible event is: 1

(a) 0 **(b)** 0.5 **(c)** 1 **(d)** −1

10. Believing that tails is ‘due’ after many heads is called: 1

(a) relative frequency **(b)** the Law of Large Numbers
(c) sampling **(d)** the Gambler's Fallacy

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. In a survey of 50 students, 10 like bananas. Find the probability that a randomly chosen student likes bananas, and the expected number among 1500 students. 2

12. A letter is picked at random from ‘PROBABILITY’. Find the probability of 2
(i) the letter B
(ii) a vowel.

13. A fair coin shows heads six times in a row. A friend says tails is sure on the next toss. Is she right? Why? 2

14. Label as impossible, less likely, even chance, more likely or certain: 2
(i) Monday follows Sunday.
(ii) It snows in Mumbai in July.

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. In a sample of 100 students, 35 like cricket. Estimate how many of 1200 students like cricket, and explain why a larger, more varied sample gives a better estimate. 3

16. In Snakes and Ladders a player has rolled three 6s in a row and says a 6 cannot come again. Find the probability of a 6 on the next roll and comment. 3

17. If $P(E) = \frac{5}{12}$, find $P(\text{not } E)$ and show that $P(E) + P(\text{not } E) = 1$. 3

OR

A letter is chosen at random from 'MATHEMATICS'. Find the probability of 3
(i) M
(ii) T
(iii) a vowel.

18. In a raffle 500 tickets are sold. Riya buys 5 tickets and one ticket wins the prize. What is the probability that Riya wins? 3

OR

Two dice are rolled. Find the probability that 3
(i) both show the same number
(ii) the sum is a prime number.

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** A die was rolled 120 times: 1→18, 2→22, 3→19, 4→21, 5→24, 6→16. 5
(i) Find the experimental probability of each face.
(ii) Find the probability of an odd number.
(iii) Compare with the theoretical value $\frac{1}{6}$ for each face.
(iv) What would you expect if the die were rolled 6000 times?

OR

(i) What is the Gambler's Fallacy? 5
(ii) A coin shows heads 6 times in a row. What is $P(\text{tails})$ on the 7th toss?
(iii) In Snakes and Ladders a player rolls three 6s in a row. What is $P(6)$ next?
(iv) Explain why the answers are not affected by earlier results.

20. Rank the following on the probability scale 0 to 1 and label each: 5
(i) The next Monday comes after Sunday.
(ii) It snows in Mumbai in July.
(iii) You roll a 3 on a fair die.
(iv) A fair coin shows heads.
(v) A bag has only red sweets and you draw a red one.

OR

Case study: A letter is picked at random from 'PROBABILITY'. 5
(i) How many letters are there?
(ii) Find $P(\text{B})$.
(iii) Find $P(\text{I})$.
(iv) Find the probability of a vowel and of a consonant, and show they add to 1.

END OF QUESTION PAPER Total: 40 Marks

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